

EXPLANATIONS

60 Advanced NEET-Pattern MCQs | Class 11 — Chapter 3 | Strictly NCERT

Each explanation references NCERT Class 11, Chapter 3 — Plant Kingdom. For quick answers only, refer to the companion Answer Key PDF.

STATEMENT-BASED QUESTIONS

Q1 - Q14

- Q 1** **Ans. (A)** Algae are grouped into Chlorophyceae (green), Phaeophyceae (brown) and Rhodophyceae (red) based primarily on the type of pigments and stored food. The pigment composition differs across the three classes — chlorophyll a and b in green, chlorophyll a, c and fucoxanthin in brown, chlorophyll a and d with phycoerythrin in red. Hence statement 4 is wrong. (NCERT XI, Ch. 3)
- Q 2** **Ans. (D)** Chlorophyceae are green algae with plant bodies that may be unicellular, colonial or filamentous, containing chlorophyll a and b. They store food as starch, not floridean starch (which is found in red algae). Examples — Volvox, Spirogyra, Chlamydomonas. So 1, 2 and 4 are correct. (NCERT XI, Ch. 3)
- Q 3** **Ans. (D)** Brown algae are mostly marine, contain chlorophyll a, c and fucoxanthin, and store food as mannitol/laminarin. The plant body is usually well-differentiated (holdfast, stipe, frond) — not unicellular. Hence 1, 2 and 3 are correct. (NCERT XI, Ch. 3)
- Q 4** **Ans. (B)** Red algae owe their colour to the r-phycoerythrin pigment that masks chlorophyll. Major pigments are chlorophyll a, d and phycoerythrin. Food reserve is floridean starch. Most red algae are marine, found in greater depths — not freshwater. So 1, 2 and 3 are correct. (NCERT XI, Ch. 3)
- Q 5** **Ans. (C)** Bryophytes are called amphibians of the plant kingdom because they need water for sexual reproduction (motile flagellated antherozoids). The gametophyte is dominant; the sporophyte is small and dependent. Bryophytes lack true vascular tissues, so statement 4 is wrong. 1, 2 and 3 are correct. (NCERT XI, Ch. 3)
- Q 6** **Ans. (A)** Bryophyte antheridia produce biflagellate antherozoids; archegonia produce one egg; fertilisation produces a zygote that develops into a multicellular sporophyte, not directly into a gametophyte. So 1, 2 and 3 are correct. (NCERT XI, Ch. 3)
- Q 7** **Ans. (A)** Pteridophytes are the first vascular plants. Sporophyte is the dominant phase. Sex organs are multicellular; spores germinate to form a prothallus (gametophyte). Pteridophytes do not form seeds — seed plants are gymnosperms and angiosperms. So 1, 2 and 3 are correct. (NCERT XI, Ch. 3)
- Q 8** **Ans. (C)** Gymnosperms are heterosporous, bearing micro- and mega-spores. Their ovules are not enclosed by an ovary — they are 'naked-seeded'. Pollination is mainly by wind. Seeds are naked, not enclosed inside fruits (fruits are a feature of angiosperms). So 1, 2 and 3 are correct. (NCERT XI, Ch. 3)

Q 9	Ans. (D) Angiosperms have seeds enclosed inside fruits; they are divided into dicotyledons and monocotyledons; the embryo sac is the female gametophyte; and they show characteristic double fertilisation. All four statements are correct. (NCERT XI, Ch. 3)
Q 10	Ans. (D) Alternation of generations is a feature of all plants (with variations in some algae). It alternates between a haploid gametophyte and a diploid sporophyte. The gametophyte produces gametes by mitosis. Spores are produced from the sporophyte by meiosis — not from the gametophyte. So 1, 2 and 3 are correct. (NCERT XI, Ch. 3)
Q 11	Ans. (A) Porphyra, Laminaria, Sargassum are eaten as food. Agar is extracted from Gelidium and Gracilaria (red algae). Algin (from brown algae) and carrageen (from red algae) are commercially used hydrocolloids. Chlorella is a unicellular alga rich in protein, used as a food supplement. All four are correct. (NCERT XI, Ch. 3)
Q 12	Ans. (B) Liverworts grow in moist shady places; their plant body is thalloid. Moss gametophyte has two stages — protonema and leafy stage. Funaria, Polytrichum and Sphagnum are common mosses. All four are correct. (NCERT XI, Ch. 3)
Q 13	Ans. (C) Pteridophytes are classified into Psilopsida (Psilotum), Lycopsida (Selaginella, Lycopodium), Sphenopsida (Equisetum) and Pteropsida (Pteris, Adiantum, Dryopteris). All four statements are correct. (NCERT XI, Ch. 3)
Q 14	Ans. (A) Selaginella and Salvinia are the classic examples of heterosporous pteridophytes. Heterospory is regarded as the precursor to seed habit. Megaspores germinate within the megasporangium, leading to the retained female gametophyte. Bryophytes are homosporous, not heterosporous. So 1, 2 and 3 are correct. (NCERT XI, Ch. 3)

ASSERTION - REASON

Q15 - Q28

Q 15	Ans. (A) Both A and R are true and R is the correct explanation. Bryophyte antherozoids are flagellated and need a film of water to swim to the egg — hence the term 'amphibians'. (NCERT XI, Ch. 3)
Q 16	Ans. (A) Both A and R are true and R is the correct explanation. The presence of vascular tissues freed pteridophytes from total dependence on water for conduction, making them the first true terrestrial plants. (NCERT XI, Ch. 3)
Q 17	Ans. (C) A is true but R is false. Gymnosperm seeds are 'naked' because their ovules are not enclosed by an ovary wall — not because they form without fertilisation. Gymnosperm seeds do form after fertilisation. (NCERT XI, Ch. 3)
Q 18	Ans. (A) Both A and R are true and R is the correct explanation. Phycoerythrin efficiently absorbs the blue-green wavelengths that reach the deeper layers of oceans, allowing red algae to photosynthesise where other algae cannot. (NCERT XI, Ch. 3)
Q 19	Ans. (A) Both A and R are true and R is the correct explanation. Gymnosperm gametophytes do not exist independently — they remain within the parent sporophyte (within the ovule and pollen grain) and are reduced compared to those of pteridophytes. (NCERT XI, Ch. 3)

Q 20	Ans. (A) Both A and R are true and R is the correct explanation. Cycas, a gymnosperm, bears naked seeds in cones (strobili) and forms neither flowers nor fruits. (NCERT XI, Ch. 3)
Q 21	Ans. (C) A is true but R is false. The gametophyte is dominant in bryophytes (including mosses). However, mosses do undergo alternation of generations — between a dominant gametophyte and a dependent sporophyte. (NCERT XI, Ch. 3)
Q 22	Ans. (D) A is false but R is true. Most pteridophytes are homosporous — only a few (e.g., Selaginella, Salvinia) are heterosporous. R correctly defines heterospory. (NCERT XI, Ch. 3)
Q 23	Ans. (B) Both A and R are true, but R is not the correct explanation. Sargassum and Laminaria are economically important chiefly as food sources and for algin production; their use as fertilizers is a separate (and minor) application, not the primary reason they are economically important. (NCERT XI, Ch. 3)
Q 24	Ans. (D) A is false but R is true. Double fertilisation is unique to angiosperms, not gymnosperms. R describes what happens in angiosperms. In gymnosperms, only one male gamete fuses with the egg. (NCERT XI, Ch. 3)
Q 25	Ans. (A) Both A and R are true and R is the correct explanation. From pteridophytes through gymnosperms to angiosperms there is a clear evolutionary trend of reduction of the gametophyte and a parallel elaboration of the sporophyte. (NCERT XI, Ch. 3)
Q 26	Ans. (C) A is true but R is false. Funaria does show a haplodiplontic life cycle (with alternation between gametophyte and sporophyte), but in Funaria the gametophyte is dominant, not the sporophyte. (NCERT XI, Ch. 3)
Q 27	Ans. (B) Both A and R are true, but R is not the correct explanation. Pinus bearing male and female cones is a feature of its reproductive structure; that Pinus is a gymnosperm with naked seeds is a separate classification fact, not the reason it has cones. (NCERT XI, Ch. 3)
Q 28	Ans. (B) Both A and R are true, but R is not the correct explanation. The reason for treating algae as polyphyletic is the diversity in their pigmentation, stored food and evolutionary origin, not merely that they are chlorophyll-bearing autotrophs. (NCERT XI, Ch. 3)

MATCH THE COLUMNS

Q29 – Q36

Q 29	Ans. (D) A–ii, B–iii, C–i, D–iv. Chlorophyceae store starch; Phaeophyceae store laminarin or mannitol; Rhodophyceae store floridean starch; Cyanobacteria store cyanophycean starch. (NCERT XI, Ch. 3)
Q 30	Ans. (A) A–iii, B–ii, C–i, D–iv. Bryophytes are amphibians of the plant kingdom; pteridophytes are the first vascular plants; gymnosperms are the first seed plants (naked seeds); angiosperms are flowering plants with seeds enclosed in fruits. (NCERT XI, Ch. 3)

Q 31	Ans. (C) A-iii, B-i, C-iv, D-ii. Psilopsida — Psilotum; Lycopsida — Selaginella, Lycopodium; Sphenopsida — Equisetum; Pteropsida — Pteris, Adiantum, Dryopteris. (NCERT XI, Ch. 3)
Q 32	Ans. (C) A-iv, B-iii, C-ii, D-i. Porphyra is eaten as food; Gelidium yields agar; Laminaria gives algin; Chlorella is rich in protein and used as a food supplement. (NCERT XI, Ch. 3)
Q 33	Ans. (D) A-iii, B-i, C-iv, D-ii. Funaria — moss; Marchantia — liverwort; Equisetum — horsetail (Sphenopsida); Selaginella — heterosporous pteridophyte. (NCERT XI, Ch. 3)
Q 34	Ans. (A) A-iv, B-iii, C-ii, D-i. Cycas shows circinate vernation of young leaves; Pinus is a conifer with male and female cones; Ginkgo is a living fossil; Wheat is a monocot angiosperm. (NCERT XI, Ch. 3)
Q 35	Ans. (B) A-iii, B-ii, C-i, D-iv. Haplontic — Volvox, Spirogyra, Chlamydomonas (sporophyte is only the zygote); Diplontic — angiosperms; Haplo-diplontic with dominant gametophyte — bryophytes; Haplo-diplontic with dominant sporophyte — pteridophytes (ferns). (NCERT XI, Ch. 3)
Q 36	Ans. (B) A-iii, B-ii, C-i, D-iv. Chlorophyceae: chl a + chl b; Phaeophyceae: chl a + chl c + fucoxanthin; Rhodophyceae: chl a + chl d + r-phycoerythrin; Cyanobacteria: chl a + phycocyanin. (NCERT XI, Ch. 3)

HOW MANY ARE CORRECT

Q37 - Q46

Q 37	Ans. (C) Bryophytes: Funaria, Marchantia, Sphagnum, Riccia (4). Selaginella and Equisetum are pteridophytes. (NCERT XI, Ch. 3)
Q 38	Ans. (A) Pteridophytes: Pteris, Adiantum, Selaginella, Equisetum (4). Cycas is a gymnosperm and Funaria is a moss (bryophyte). (NCERT XI, Ch. 3)
Q 39	Ans. (C) Gymnosperms: Cycas, Pinus, Ginkgo, Cedrus (4). Mango is an angiosperm and Equisetum is a pteridophyte. (NCERT XI, Ch. 3)
Q 40	Ans. (C) Correct: 1, 2 and 3 (3 correct). Bryophytes lack true vascular roots/stems/leaves, have flagellated antherozoids, and have a dominant gametophyte. They do not produce seeds. (NCERT XI, Ch. 3)
Q 41	Ans. (A) Correct features: 1, 2 and 4 — 3 correct. Algal sex organs are not surrounded by a multicellular jacket (that's a bryophyte/pteridophyte feature); and an embryo is not formed after fertilisation in algae — the zygote develops directly into the new plant. (NCERT XI, Ch. 3)
Q 42	Ans. (D) Correct: 1, 2 and 4 — 3 correct. Gymnosperm seeds are not enclosed in a fruit (statement 3 wrong). The female gametophyte is multicellular but retained within the megasporangium — not free-living (statement 5 wrong). (NCERT XI, Ch. 3)
Q 43	Ans. (B) Features unique to angiosperms: seeds in fruits, double fertilisation and triploid endosperm — 3 unique. Naked ovules are a gymnosperm feature; heterospory occurs in some pteridophytes and all gymnosperms too. (NCERT XI, Ch. 3)

- Q 44** Ans. (D) Correctly matched: Volvox, Sargassum, Polysiphonia and Equisetum — 4. Funaria is a moss, not a liverwort. (NCERT XI, Ch. 3)
- Q 45** Ans. (B) Haplo-diplontic: Moss, Fern and Selaginella (3). Spirogyra and Volvox are haplontic green algae. (NCERT XI, Ch. 3)
- Q 46** Ans. (D) Incorrect statements: 1 (algae include multicellular forms too), 3 (Selaginella is heterosporous) and 4 (gymnosperms do not produce flowers). 3 incorrect. Statements 2 and 5 are correct. (NCERT XI, Ch. 3)

CHOOSE INCORRECT / EXCEPT

Q47 - Q54

- Q 47** Ans. (B) Phaeophyceae are predominantly marine, not freshwater. The other three statements are correct. (NCERT XI, Ch. 3)
- Q 48** Ans. (D) Funaria, Marchantia and Sphagnum are bryophytes. Selaginella is a heterosporous pteridophyte. (NCERT XI, Ch. 3)
- Q 49** Ans. (D) Pteridophytes do not form seeds at all, let alone fruits. The other features are correct. (NCERT XI, Ch. 3)
- Q 50** Ans. (C) Gymnosperm female gametophytes are not free-living — they are retained within the megasporangium and are not the dominant phase. The other statements are correct. (NCERT XI, Ch. 3)
- Q 51** Ans. (C) Pteris, Adiantum and Dryopteris are Pteropsida. Equisetum belongs to Sphenopsida. (NCERT XI, Ch. 3)
- Q 52** Ans. (B) In algae, the zygote develops directly into a new plant — there is no embryo stage. The other statements are correct. (NCERT XI, Ch. 3)
- Q 53** Ans. (D) Sphagnum is a moss, not a liverwort. The other three pairs are correctly matched. (NCERT XI, Ch. 3)
- Q 54** Ans. (B) In angiosperms, ovules are enclosed inside an ovary — that is the defining feature of the group. Naked ovules are a gymnosperm feature. (NCERT XI, Ch. 3)

STANDARD NEET MCQs

Q55 - Q60

- Q 55** Ans. (C) Bryophytes are called amphibians of the plant kingdom because they live on land but need water for sexual reproduction. (NCERT XI, Ch. 3)
- Q 56** Ans. (B) Pteridophytes are the first plants to possess true vascular tissues — xylem and phloem. (NCERT XI, Ch. 3)
- Q 57** Ans. (B) In bryophytes, the dominant photosynthetic phase is the multicellular gametophyte; the sporophyte is small and dependent on the gametophyte. (NCERT XI, Ch. 3)

- Q 58** **Ans. (D)** r-Phycoerythrin masks the chlorophyll pigments and imparts the characteristic red colour to Rhodophyceae. (NCERT XI, Ch. 3)
- Q 59** **Ans. (B)** Most pteridophytes are homosporous. Selaginella and Salvinia are exceptional heterosporous pteridophytes. Gymnosperms are uniformly heterosporous. (NCERT XI, Ch. 3)
- Q 60** **Ans. (C)** Double fertilisation is unique to angiosperms — one male gamete fuses with the egg, the other with the polar nuclei, producing a diploid zygote and triploid endosperm. (NCERT XI, Ch. 3)